Florida State University
Department of Scientific Computing

Written Preliminary Examination

Summer 2024

Question - 5

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Contents

1 Approximation and Differentiation[Dr. Huang]

1.1 Part (a)

The given objective function is:

$$W = \int_{-1}^{1} [f(x) - \sum_{j=0}^{n} c_j P_j(x)]^2 \, dx$$

Differentiating W with respect to c_i:

$$\begin{aligned}
\frac{\partial W}{\partial c_i} &= \frac{\partial}{\partial c_i} \int_{-1}^{1} [f(x) - \sum_{j=0}^{n} c_j P_j(x)]^2 \, dx \\
&= \int_{-1}^{1} \frac{\partial}{\partial c_i} [f(x) - \sum_{j=0}^{n} c_j P_j(x)]^2 \, dx \quad \text{(Leibniz rule for differentiation under the integral sign)} \\
&= 2 \int_{-1}^{1} [f(x) - \sum_{j=0}^{n} c_j P_j(x)] \frac{\partial}{\partial c_i} [f(x) - \sum_{j=0}^{n} c_j P_j(x)] \, dx \quad \text{(Using chain rule.)} \\
&= -2 \int_{-1}^{1} \left[f(x) - \sum_{j=0}^{n} c_j P_j(x) \right] P_i(x) \, dx \quad \text{(Only } P_j = P_i \text{ survived)} \\
&= -2 \int_{-1}^{1} \left[f(x) P_i(x) - \left(\sum_{j=0}^{n} c_j P_j(x) \right) P_i(x) \right] \, dx
\end{aligned}$$

Hence, we have:

$$\boxed{\frac{\partial W}{\partial c_i} = -2 \int_{-1}^{1} \left[f(x) P_i(x) - \left(\sum_{j=0}^{n} c_j P_j(x) \right) P_i(x) \right] dx}$$

1.2 Part (b)

To derive the expression for c_i, we will use the fact that Legendre polynomials of different degrees are orthogonal over $[-1, 1]$ i.e.,

$$\int_{-1}^{1} P_n(x) P_m(x) dx = \frac{2\delta_{mn}}{2n+1} \quad \text{where} \quad \delta_{mn} = \begin{cases} 1 & \text{if } m = n, \\ 0 & \text{if } m \neq n. \end{cases}$$

We will use this in the expression obtained in part (a).

$$\begin{aligned}
\frac{\partial W}{\partial c_i} &= -2 \left[\int_{-1}^{1} f(x) P_i(x) \, dx - \int_{-1}^{1} \left(\sum_{j=0}^{n} c_j P_j(x) \right) P_i(x) \, dx \right] \\
&= -2 \left[\int_{-1}^{1} f(x) P_i(x) \, dx - \frac{2c_i}{2i+1} \right]
\end{aligned}$$

So, the final expression for $\frac{\partial W}{\partial c_i}$ is:

$$\boxed{\frac{\partial W}{\partial c_i} = -2 \left[\int_{-1}^{1} f(x) P_i(x) \, dx - \frac{2c_i}{2i+1} \right]}$$

To solve for c_i, we can put this derivative equal to zero:

$$\frac{\partial W}{\partial c_i} = -2\left[\int_{-1}^{1} f(x)P_i(x)\,dx - \frac{2c_i}{2i+1}\right] = 0$$

$$\implies \int_{-1}^{1} f(x)P_i(x)\,dx = \frac{2c_i}{2i+1}$$

Therefore, the expression for c_i is:

$$\boxed{c_i = \frac{(2i+1)}{2}\int_{-1}^{1} f(x)P_i(x)\,dx}$$

1.3 Part (c)

Given function to be approximated is,

$$f(x) = e^{-x^2}\sin(x-1) + 1$$

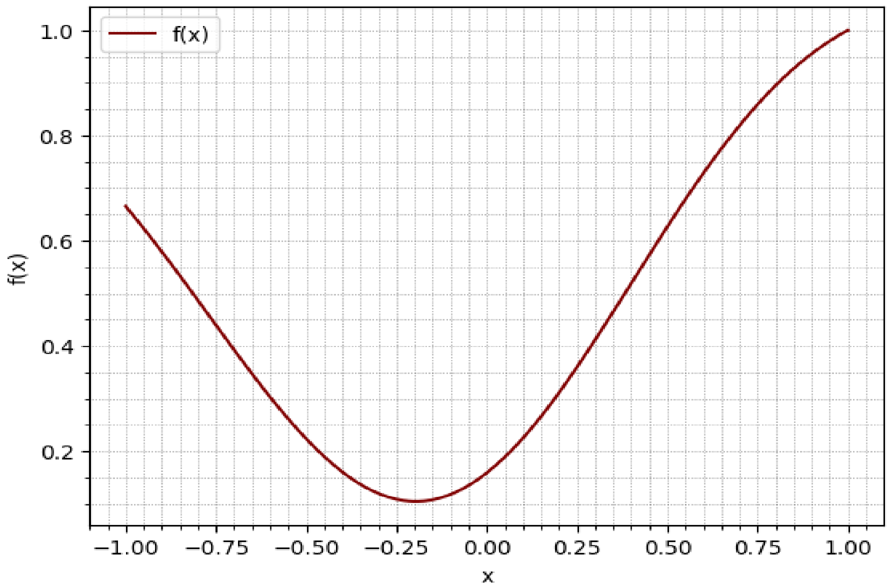

Figure 1: Original function f(x).

Calculation of coefficient c_i:

```
==============================================
Coefficient c0: 0.44784831278662507
Coefficient c1: 0.28091490992363866
Coefficient c2: 0.4838938841641618
Coefficient c3: -0.13645674773391614
Coefficient c4: -0.11273025713192437
==============================================
```

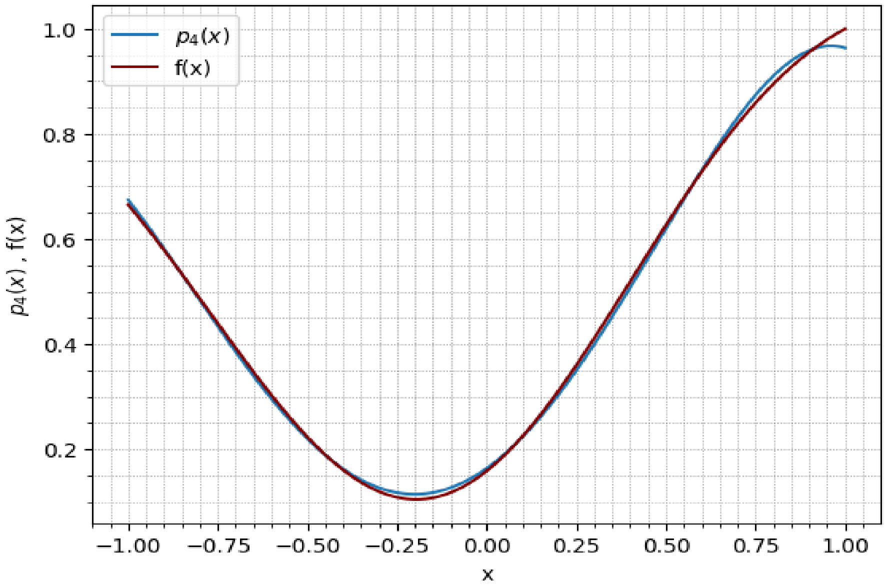

Figure 2: Comparison of f(x) and approximation using Legendre Polynomial.

1.4 Part (d)

Calculation of coefficient c_i:

```
===================================
c0: 0.4478686737695029
c1: 0.2809144444406612
c2: 0.4844153372676482
c3: -0.13626722666974048
c4: -0.10999294494482652
===================================
```

The resulting coefficient obtained by minimizing W are matching to those obtained in part (c). Only c_4 is slightly different.

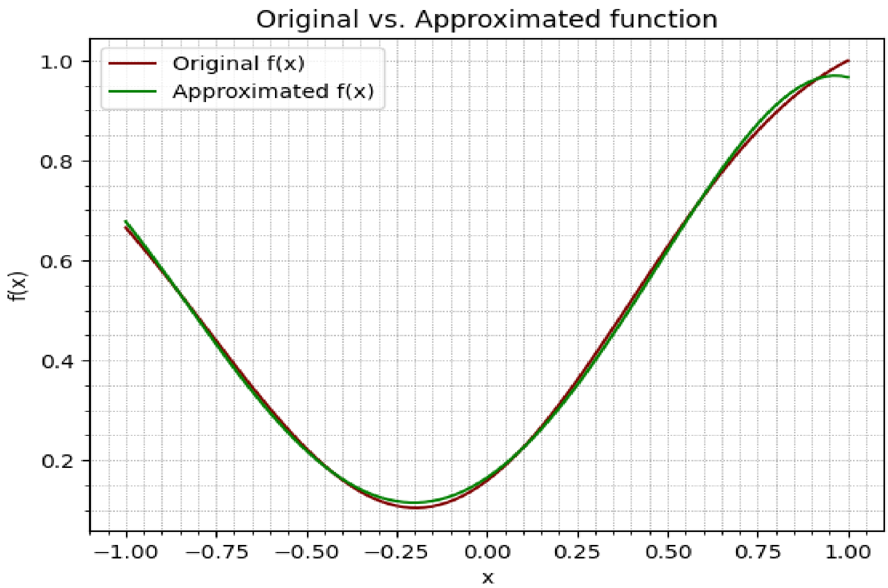

Figure 3: Comparison of f(x) and approximated function using Optimization.

1.5 Part (e)

I was getting some warnings in my Python code for this part. I tried a lot and couldn't fixed it so I used Matlab for doing this part.

Calculation of coefficient c_i using complex step differntiation:

```
===================================
Optimized coefficients c_i:
c0 = 0.44785
c1 = 0.28092
c2 = 0.48389
c3 = -0.13645
c4 = -0.11273
===================================
```

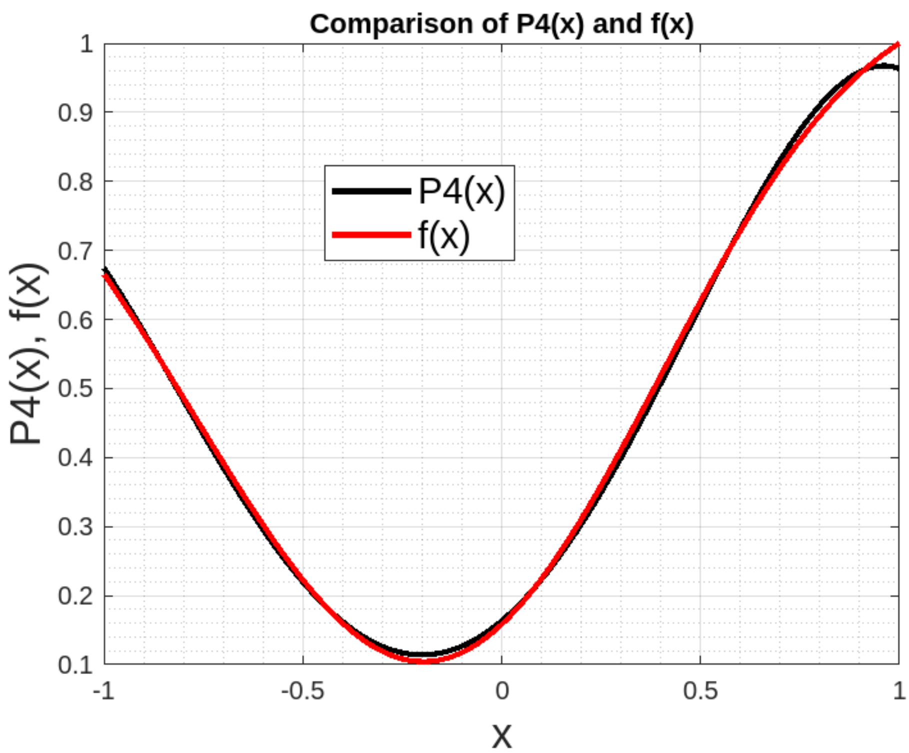

Figure 4: Comparison of approximated function and original function.

NOTES:

- I have used `scipy.integrate.quad` for integration.
- For optimization (part d), `BFGS` method from `scipy.optimize` is used.
- For part (e), the Matlab code is used and file name is *Q5_e.m*

References

- ACSI Lecture Notes and codes, Dr. Huang
- https://math.stackexchange.com/questions/2147167/